

Homework 10

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1. Let $c_0 = \{(x_1, x_2, \dots) \in \ell^\infty : \lim_{n \rightarrow \infty} x_n = 0\}$. Show that $(c_0)'$ is isomorphic to ℓ^1 .

Proof. Given any $a = (a_n) \in \ell^1$, we have a linear functional f_a which maps $x = (x_n) \in c_0$ by

$$f_a(x) = \sum_{i=1}^{\infty} a_i x_i,$$

which is bounded since $|\sum_{i=1}^{\infty} a_i x_i| \leq \|x\|_{\infty} \sum_{i=1}^{\infty} |a_i| = \|x\|_{\infty} \|a\|_1$. Now define e_i to be the sequence with 1 in the i^{th} position and 0 elsewhere. Then given a bounded linear functional $f \in (c_0)'$, we can define a sequence $a = (f(e_n))_{n \in \mathbb{Z}}$. Then given any $x = (x_1, x_2, \dots) \in c_0$,

$$\begin{aligned} f_a(x) &= \sum_{i=1}^{\infty} a_i x_i \\ &= \sum_{i=1}^{\infty} f(e_i) x_i \\ &= \sum_{i=1}^{\infty} f(x_i e_i) \\ &= f\left(\sum_{i=1}^{\infty} x_i e_i\right) \\ &= f(x), \end{aligned}$$

so f is indeed realized by this mapping of elements of ℓ^1 to functionals on c_0 . □

2. In an inner product space, we say that x, y are orthogonal (sometimes we write $x \perp y$) if $\langle x, y \rangle = 0$. Show that if x, y are non-zero and orthogonal then they must be linearly independent. Extend this result to an arbitrary finite number of pairwise orthogonal vectors and prove it.

Proof. I will just prove the general result since it includes the two-vector result. Suppose we have nonzero vectors $x_1, \dots, x_n$ which are all pairwise orthogonal. Then we want to show that

$$c_1 x_1 + \dots + c_n x_n = 0 \tag{1}$$

implies $c_1 = \dots = c_n = 0$. Indeed, assume we have (1) for some scalars $c_1, \dots, c_n$. Then for each $i \in \{1, \dots, n\}$,

$$\begin{aligned} 0 &= \langle c_1 x_1 + \dots + c_n x_n, x_i \rangle \\ &= c_1 \langle x_1, x_i \rangle + \dots + c_n \langle x_n, x_i \rangle \\ &= c_i \langle x_i, x_i \rangle \end{aligned}$$

since $\langle x_j, x_i \rangle = 0$ whenever $i \neq j$ by hypothesis. Since x_i is nonzero, $\langle x_i, x_i \rangle > 0$, so the final equality implies $c_i = 0$. Our choice of i was arbitrary, so we are done. $\square$

3. Show that if x_n converges to x and y_n converges to y with respect to the induced norm then $\langle x_n, y_n \rangle$ converges to $\langle x, y \rangle$.

Proof. First, observe that

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n, y_n \rangle - \langle x, y_n \rangle + \langle x, y_n \rangle - \langle x, y \rangle| \\ &= \langle x_n - x, y_n \rangle + \langle x, y_n - y \rangle \\ &\leq \|x_n - x\| \cdot \|y_n\| + \|x\| \cdot \|y_n - y\|. \end{aligned}$$

We will study the final equation to get our result. The second term goes to zero because $\|x\|$ is constant with respect to n , and $\|y_n - y\|$ goes to zero as $n \rightarrow \infty$ since $y_n \rightarrow y$. The first term has $\|x_n - x\| \rightarrow 0$ for the same reason, and of course $y_n \rightarrow y$, so that first term will also tend to 0 as $n \rightarrow \infty$. Thus we get that the entire RHS of that inequality goes to 0 as $n \rightarrow \infty$, and thus $|\langle x_n, y_n \rangle - \langle x, y \rangle|$ does as well since it is bounded above by 0. $\square$

4. Let $T : X \rightarrow X$ be a bounded linear operator on a complex inner product space X . If $\langle T(x), x \rangle = 0$ for all $x \in X$, show that $T = 0$. Also, give an example where this is false in real inner product spaces.

Proof. Let $a \in \mathbb{C}$ and $u, v \in X$. Then

$$\begin{aligned} 0 &= \langle T(u + av), u + av \rangle \\ &= \langle T(u) + aT(v), u + av \rangle \\ &= \langle T(u), u \rangle + \bar{a} \langle T(u), v \rangle + a \langle T(v), u \rangle + a\bar{a} \langle T(v), v \rangle \\ &= \bar{a} \langle T(u), v \rangle + a \langle T(v), u \rangle. \end{aligned}$$

Now since a is arbitrary, we can look at a couple values of a in particular. Let $A = \langle T(u), v \rangle$ and $B = \langle T(v), u \rangle$ for notational ease. Then when $a = 1$ we have

$$A + B = 0$$

and when $a = i$ we have

$$-iA + iB = 0 \implies -A + B = 0.$$

Adding the two equations together, we get that $2B = 0$, so $B = 0$, and then plugging that into the first equation we get $A = 0$. Therefore $\langle T(u), v \rangle = 0$ for all $u, v \in X$. If we choose $v = T(u)$, this yields $\langle T(u), T(u) \rangle = 0$, which is true only when $T(u) = 0$. This is true for all u , so $T = 0$.

For the real counterexample, choose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (-y, x)$. Then indeed for any $(x, y) \in \mathbb{R}^2$, we have

$$\begin{aligned}\langle T(x, y), (x, y) \rangle &= \langle (-y, x), (x, y) \rangle \\ &= -xy + xy \\ &= 0.\end{aligned}$$

Thus T satisfies the criterion, but it is clearly not the zero map (e.g. $(1, 0) \mapsto (0, 1)$). □